SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 50

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Code of Conduct

I M. Greeshma / 192524195 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Greeshma

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

cloud computing fundamentals

Characteristics

- Elasticity
- scalability
- Resource pooling
- on-demand access
- Measured service

Benefits

- low cost
- flexibility
- Easy access
- Backup

Cloud Computing

delivering computing services over internet

Limitations

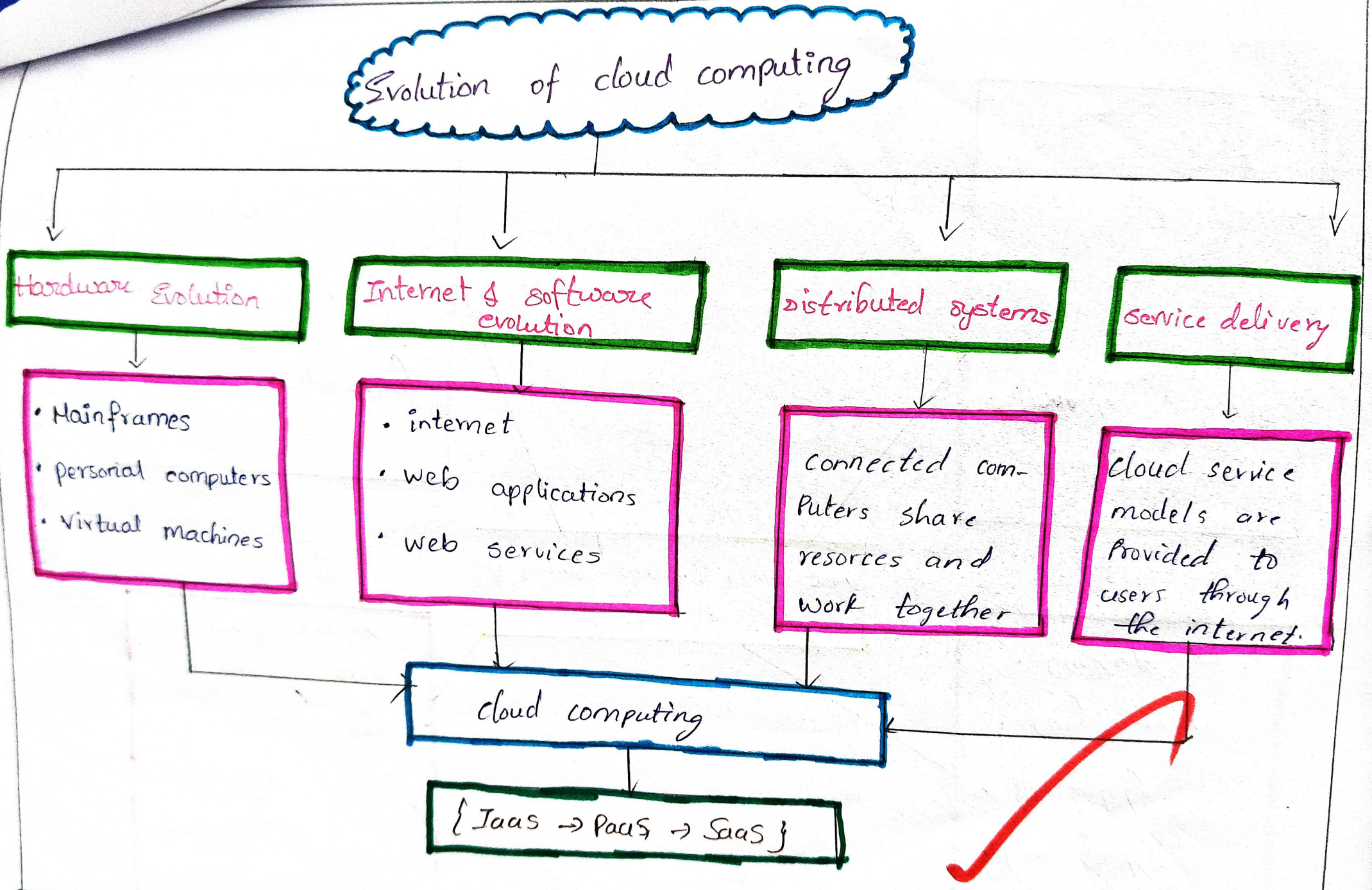
- internet dependency
- securify risks
- downtime

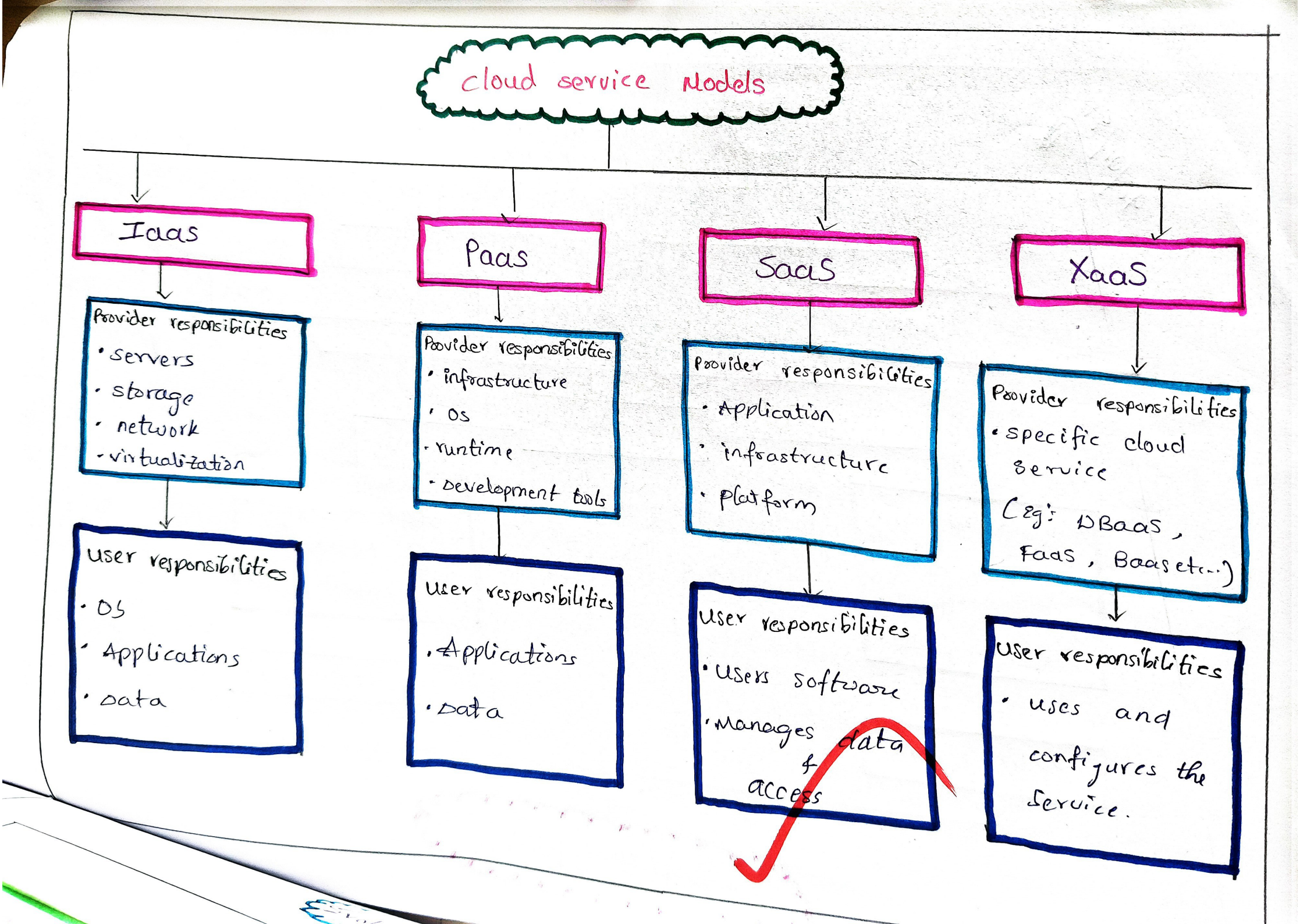
Examples

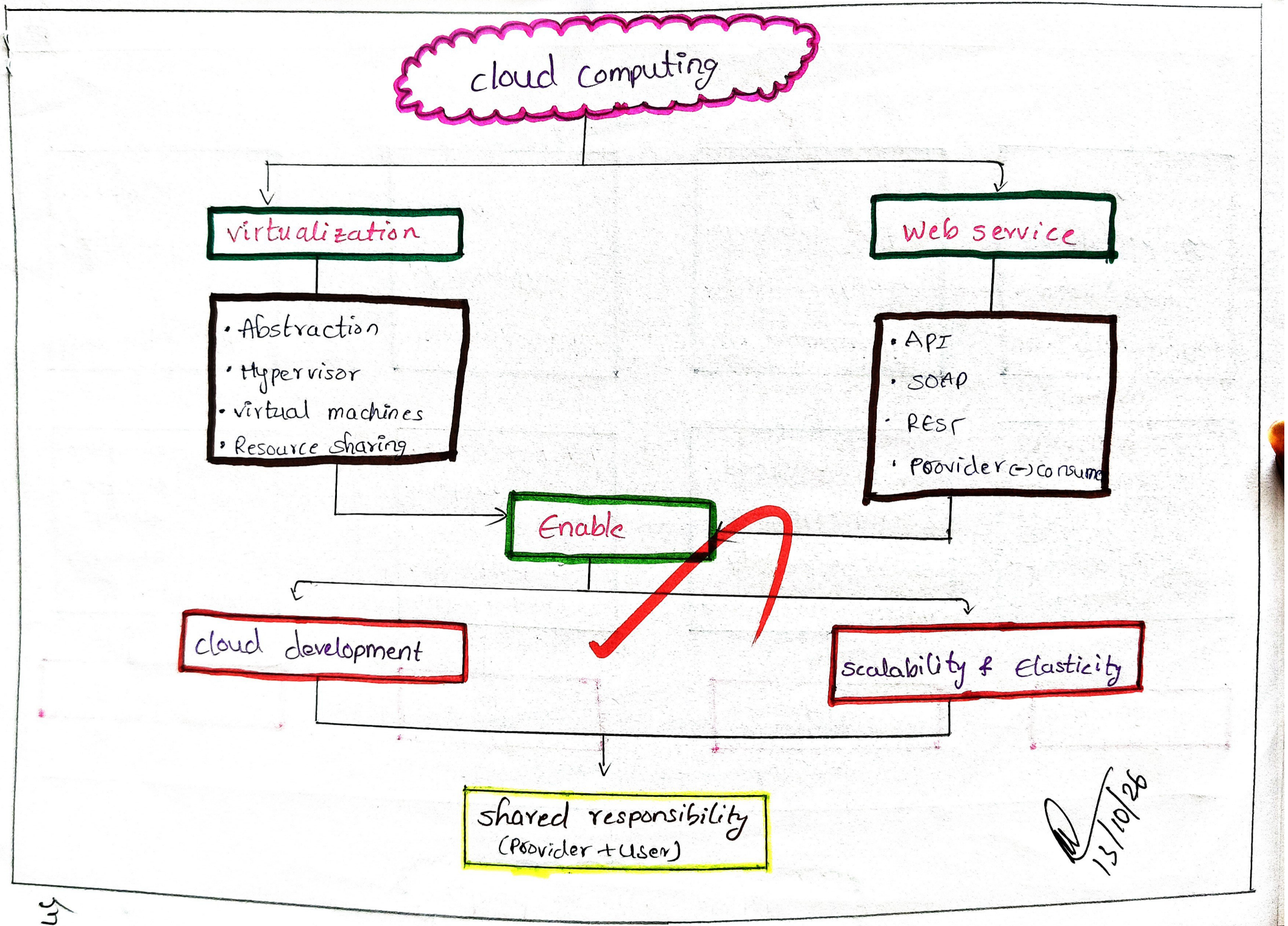
- Google drive
- AWS
- Microsoft Azure



Evolution of cloud computing




cloud service Models
Iaas
Provider responsibilities
• servers
• storage
• network
• virtualization
User responsibilities
• OS
• Applications
• Data
Paas
Provider responsibilities
• infrastructure
• OS
• runtime
• development tools
User responsibilities
• Applications
• Data
SaaS
Provider responsibilities
• Application
• infrastructure
• platform
User responsibilities
• Users software
• Manages data & access
XaaS
Provider responsibilities
• specific cloud service
(Eg: DBaaS, Faas, Baas etc..)
User responsibilities
• uses and configures the service.


cloud computing
virtualization
Web service
• Abstraction
• Hypervisor
• virtual machines
• Resource sharing
• API
• SOAP
• REST
• Provider → consume
Enable
cloud development
scalability & Elasticity
shared responsibility
(Provider + User)
13/10/26